

PE-EC801B — Fiber Optic Communication

Semester VIII • 3L:0T:0P • 3 Credits

Module 1 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Vector Nature of Light — EM Wave Fundamentals

Light as an Electromagnetic Wave

Light is a **transverse electromagnetic (TEM) wave** in which the electric field $\vec{E}$ and magnetic field $\vec{H}$ are perpendicular to each other and to the direction of propagation $\hat{k}$. In free space, both fields satisfy Maxwell's wave equation.

1.1 Maxwell's Equations and Wave Equation

For a linear, isotropic, source-free dielectric medium (permittivity ϵ , permeability μ), Maxwell's curl equations are:

$$\nabla \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}, \quad \nabla \times \vec{H} = \epsilon \frac{\partial \vec{E}}{\partial t}$$

Taking the curl of the first equation and using the second gives the vector wave equation:

$$\nabla^2 \vec{E} = \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

The speed of light in the medium is $v = 1/\sqrt{\mu\epsilon}$. In free space, $v = c = 3 \times 10^8$ m/s, where $c = 1/\sqrt{\mu_0\epsilon_0}$.

1.2 E and H Field Relationship

For a plane wave propagating in the $+z$ direction with wavenumber $k = \omega\sqrt{\mu\epsilon}$:

$$\vec{E} = E_0 e^{j(\omega t - kz)} \hat{x}, \quad \vec{H} = \frac{E_0}{\eta} e^{j(\omega t - kz)} \hat{y}$$

where $\eta = \sqrt{\mu/\epsilon}$ is the intrinsic impedance of the medium. In free space, $\eta_0 = 377 \Omega$.

Key EM Wave Parameters

Wave number	$k = \frac{2\pi}{\lambda} = \frac{\omega}{v} = \omega\sqrt{\mu\epsilon}$
Refractive index	$n = \frac{c}{v} = \sqrt{\mu_r\epsilon_r} \approx \sqrt{\epsilon_r}$ (non-magnetic)
Intrinsic impedance	$\eta = \sqrt{\mu/\epsilon}$; free space $\eta_0 = 377 \Omega$
Poynting vector	$\vec{S} = \vec{E} \times \vec{H}$ (power flow direction)

2. Propagation of Light — Reflection, Refraction, and TIR

Snell's Law and Critical Angle

When light travels from medium 1 (refractive index n_1) to medium 2 (n_2), **Snell's law** governs the angle of refraction: $n_1 \sin \theta_1 = n_2 \sin \theta_2$. The **critical angle** θ_c is the angle of incidence in medium 1 at which the refracted ray in medium 2 travels along the interface ($\theta_2 = 90^\circ$). **Total Internal Reflection (TIR)** occurs for $\theta_1 > \theta_c$ when $n_1 > n_2$.

Snell's Law and Critical Angle Formulas

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad (\text{Snell's Law})$$

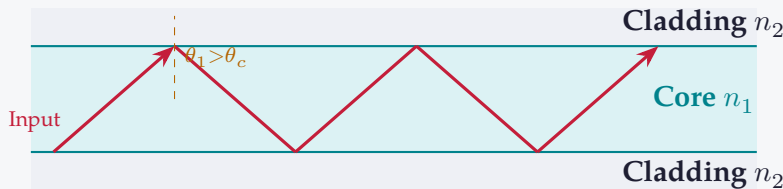
$$\sin \theta_c = \frac{n_2}{n_1} \quad (n_1 > n_2) \Rightarrow \theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right)$$

TIR condition: $\theta_1 \geq \theta_c$ (light stays in the denser medium).

2.1 Propagation in a Cylindrical Dielectric Rod (Optical Fiber)

An optical fiber is a cylindrical dielectric rod comprising a **core** (refractive index n_1) surrounded by a **cladding** (refractive index $n_2 < n_1$). Light injected at the core-cladding interface at an angle $\geq \theta_c$ undergoes TIR repeatedly and is guided along the fiber without escaping.

TIR in an Optical Fiber — Ray Path



3. Ray Model — Acceptance Angle and Numerical Aperture

Numerical Aperture (NA) and Acceptance Angle

The **acceptance angle** θ_a (or half-cone angle) is the maximum angle at which light may enter the fiber end-face and still undergo TIR. The **Numerical Aperture (NA)** is a dimensionless measure of the light-gathering ability of the fiber: $NA = n_0 \sin \theta_a$ where n_0 is the refractive index of the medium at the input (usually air, $n_0 = 1$).

3.1 Derivation of NA

Consider a ray entering the flat end of the fiber at angle θ_i (in air). By Snell's law at the end-face:

$$\sin \theta_i = n_1 \sin \theta_r$$

where θ_r is the refraction angle inside the core. For TIR at the core–cladding interface, the ray must strike that interface at an angle $\geq \theta_c$, meaning the complementary angle $\theta_r \leq (90^\circ - \theta_c)$. Using $\sin \theta_c = n_2/n_1$:

Numerical Aperture — Complete Formula

$$\text{NA} = \sin \theta_a = \sqrt{n_1^2 - n_2^2}$$

$$\text{NA} = n_1 \sqrt{2\Delta}, \quad \Delta = \frac{n_1^2 - n_2^2}{2n_1^2} \approx \frac{n_1 - n_2}{n_1} \quad (\text{relative refractive index difference})$$

Acceptance angle: $\theta_a = \sin^{-1}(\text{NA})$ (full acceptance cone = $2\theta_a$).

3.2 Meridional vs Skew Rays

- **Meridional rays:** Rays that pass through the fiber axis. Their path lies entirely in a plane containing the axis. They follow simple zigzag trajectories and are the easiest to analyze.
- **Skew rays:** Rays that do not pass through the fiber axis. They travel in a helical path around the axis. They have a larger acceptance angle than meridional rays and carry higher-order modes.

Worked Example: Calculate NA and Acceptance Angle

Given: $n_1 = 1.48$, $n_2 = 1.46$. Find NA and θ_a .

$$\text{NA} = \sqrt{1.48^2 - 1.46^2} = \sqrt{2.1904 - 2.1316} = \sqrt{0.0588} \approx 0.2425$$

$$\theta_a = \sin^{-1}(0.2425) \approx 14.03^\circ, \quad \Delta = \frac{1.48 - 1.46}{1.48} \approx 1.35\%$$

4. Wave Model — Modes and V-Number

Modes in an Optical Fiber

A **mode** is a distinct spatial distribution (pattern) of the electromagnetic field that can propagate along the fiber without changing shape. Only certain discrete field patterns satisfy the boundary conditions at the core–cladding interface; these are the allowed or guided modes.

4.1 Normalized Frequency (V-Number)

The **V-number** (also called the normalized frequency or normalized waveguide parameter) determines how many modes a fiber can support.

V-Number Formula

$$V = \frac{2\pi a}{\lambda} \text{NA} = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2} = \frac{2\pi a}{\lambda} n_1 \sqrt{2\Delta}$$

where a = core radius, λ = free-space wavelength of light.

- $V < 2.405 \Rightarrow$ **single-mode** fiber (only the HE_{11} fundamental mode propagates).
- $V \geq 2.405 \Rightarrow$ **multimode** fiber.
- Total number of modes (step-index): $M \approx \frac{V^2}{2}$.
- Total number of modes (graded-index, parabolic): $M \approx \frac{V^2}{4}$.

4.2 Cutoff Condition and Mode Designation

Modes are designated HE_{mn} or EH_{mn} (hybrid modes), TE_{0n} , TM_{0n} . The fundamental mode HE_{11} has no cutoff. The first higher-order mode TE_{01} cuts off at $V = 2.405$ (first zero of Bessel function J_0).

4.3 Modal Analysis of a Step-Index Fiber

The electromagnetic field inside the core satisfies the wave equation in cylindrical coordinates, yielding Bessel functions J_n (oscillatory, inside core) and modified Bessel functions K_n (decaying, inside cladding). Applying continuity of tangential E and H at the core-cladding boundary gives the **characteristic equation** (eigenvalue equation) for the allowed propagation constants β :

Characteristic Equation for Step-Index Fiber

$$\left[\frac{J'_n(U)}{U J_n(U)} + \frac{K'_n(W)}{W K_n(W)} \right] \left[\frac{n_1^2 J'_n(U)}{U J_n(U)} + \frac{n_2^2 K'_n(W)}{W K_n(W)} \right] = n^2 \left(\frac{1}{U^2} + \frac{1}{W^2} \right)^2$$

where the normalized transverse wave numbers are:

$$U = a \sqrt{n_1^2 k_0^2 - \beta^2}, \quad W = a \sqrt{\beta^2 - n_2^2 k_0^2}, \quad U^2 + W^2 = V^2$$

For the **LP_{01} (HE_{11}) fundamental mode**: the simplified characteristic equation is $J_0(U)K_1(W) + J_1(U)K_0(W) = 0$. This transcendental equation is solved numerically to find β for each mode.

Key results from modal analysis:

- Each solution β_m corresponds to one guided mode with phase velocity $v_{ph} = \omega/\beta_m$.
- Group velocity: $v_g = d\omega/d\beta$ (determines pulse propagation and intermodal dispersion).
- For LP modes: LP_{lm} modes correspond to $HE_{l+1,m}$ and $EH_{l-1,m}$ modes. The LP approximation ($n_1 \approx n_2$) simplifies the characteristic equation to $J_{l-1}(U)/[U J_l(U)] = -K_{l-1}(W)/[W K_l(W)]$.

Worked Example: V-Number and Mode Count

Given: $a = 25 \mu\text{m}$, $\lambda = 1300 \text{ nm}$, $n_1 = 1.48$, $n_2 = 1.46$.

$$\text{NA} = 0.2425 \quad (\text{from previous example})$$

$$V = \frac{2\pi \times 25 \times 10^{-6}}{1300 \times 10^{-9}} \times 0.2425 = \frac{2\pi \times 25}{1.3} \times 0.2425 \approx 29.2$$

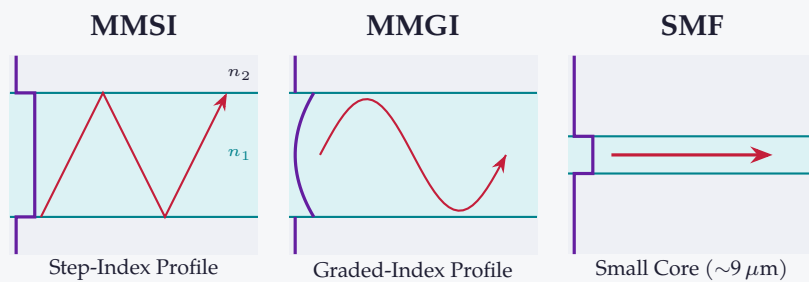
Number of modes $M \approx V^2/2 \approx 426$. Since $V \gg 2.405$, this is a multimode fiber.

5. Types of Optical Fibers

Classification of Optical Fibers

Optical fibers are classified by (a) **index profile** of the core — step-index (SI) or graded-index (GI), and (b) **number of modes** — single-mode (SM) or multimode (MM). The three main types are: Single-Mode Step-Index (SMF), Multimode Step-Index (MMSI), and Multimode Graded-Index (MMGI).

Refractive Index Profiles and Ray Paths — Three Fiber Types



5.1 Comparison Table — SMF vs MMSI vs MMGI

Parameter	SMF	MMSI	MMGI
Core diameter	$\sim 8\text{--}10\ \mu\text{m}$	$\sim 50\text{--}200\ \mu\text{m}$	$\sim 50\text{--}62.5\ \mu\text{m}$
Cladding diameter	$125\ \mu\text{m}$	$125\text{--}400\ \mu\text{m}$	$125\ \mu\text{m}$
Index profile	Step	Step	Parabolic ($\alpha \approx 2$)
V-number	< 2.405	$\gg 2.405$	$\gg 2.405$
No. of modes	1 (HE_{11} only)	$\sim V^2/2$ (hundreds)	$\sim V^2/4$
Bandwidth	Very high ($>10\ \text{GHz}\cdot\text{km}$)	Low ($\sim 10\text{--}25\ \text{MHz}\cdot\text{km}$)	Medium ($\sim 400\text{--}2000\ \text{MHz}\cdot\text{km}$)
Modal dispersion	None	High	Low (near-zero for $\alpha=2$)
NA	$\sim 0.1\text{--}0.12$	$\sim 0.2\text{--}0.5$	$\sim 0.2\text{--}0.3$
Coupling efficiency	Low (small core)	High	Medium
Light source	Laser only	LED or Laser	LED or Laser
Application	Long-haul, WDM	Short-haul, LAN	LAN, medium distance

Quick Revision — Module 1

Light Propagation and Optical Fiber Fundamentals | PE-EC801B

Key Formulas at a Glance

Snell's Law	$n_1 \sin \theta_1 = n_2 \sin \theta_2$
Critical angle	$\theta_c = \sin^{-1}(n_2/n_1) \quad (n_1 > n_2)$
Numerical Aperture	$NA = \sqrt{n_1^2 - n_2^2} = n_1 \sqrt{2\Delta}$
Relative index diff.	$\Delta = (n_1 - n_2)/n_1$ (approx. for $n_1 \approx n_2$)
V-number	$V = (2\pi a/\lambda) NA$
SMF condition	$V < 2.405$
Mode count (MMSI)	$M \approx V^2/2$
Mode count (MMGI)	$M \approx V^2/4$
Wave speed in medium	$v = c/n; \quad n = \sqrt{\epsilon_r}$

One-Line Definitions

- **TIR:** Total Internal Reflection — light stays in denser medium when $\theta_i \geq \theta_c$.
- **NA:** Numerical Aperture — measure of light-gathering ability of the fiber.
- **V-number:** Normalized frequency parameter determining number of guided modes.
- **Meridional ray:** Ray passing through the fiber axis, zigzag path.
- **Skew ray:** Ray traveling in a helical path, does not pass through axis.
- **Mode:** Distinct stable field distribution pattern guided along the fiber.
- **SMF:** Single-mode fiber with $V < 2.405$; eliminates modal dispersion.
- **MMGI:** Multimode graded-index fiber; parabolic profile minimizes modal dispersion.

Frequently Asked / Viva Questions

Q1. What is Total Internal Reflection and when does it occur?

TIR occurs when light travels from a denser medium (n_1) to a rarer medium ($n_2 < n_1$) and the angle of incidence exceeds the critical angle $\theta_c = \sin^{-1}(n_2/n_1)$. All incident light is reflected back; no energy enters medium 2.

Q2. Why is the refractive index of the core greater than the cladding in an optical fiber?

TIR — which is the guiding mechanism — requires light to travel from a denser

Frequently Asked / Viva Questions (contd.)

to rarer medium. Thus $n_1 > n_2$ is essential for guided propagation.

Q3. Define Numerical Aperture (NA). What does a higher NA imply?

$NA = \sqrt{n_1^2 - n_2^2}$ is the sine of the maximum acceptance angle. Higher NA means the fiber accepts light over a wider cone but also supports more modes (greater modal dispersion).

Q4. What is the V-number and what is its significance?

The V-number $= (2\pi a/\lambda) NA$ is the normalized frequency. It determines the number of guided modes. For $V < 2.405$, only one mode propagates (single-mode operation).

Q5. Distinguish between meridional and skew rays.

Meridional rays pass through the fiber axis and travel in a plane containing it. Skew rays do not intersect the axis; they travel helically. Skew rays have a slightly larger effective acceptance angle.

Q6. Why does a graded-index fiber have lower modal dispersion than step-index?

In MMGI, rays in the outer (faster) region travel longer paths but at higher speeds (lower n), while rays near the axis travel slower but shorter paths. This self-equalizing effect reduces pulse spreading.

Q7. What is the cutoff condition for single-mode operation?

$V < 2.405$, where 2.405 is the first zero of the Bessel function J_0 . Below this, only the fundamental HE_{11} mode propagates.

Q8. What is Δ (relative refractive index difference)?

$\Delta = (n_1 - n_2)/n_1 \approx (n_1^2 - n_2^2)/(2n_1^2)$. It is typically 0.1–2% for communication fibers. Larger Δ means higher NA but also higher dispersion.

Q9. How does the number of modes depend on V-number?

For step-index: $M \approx V^2/2$. For graded-index (parabolic): $M \approx V^2/4$. The graded profile halves the mode count for the same V-number.

Q10. What are the dimensions of a standard SMF and MMSI fiber?

SMF: core $\sim 8\text{--}10\ \mu\text{m}$, cladding $125\ \mu\text{m}$. MMSI: core $50\text{--}200\ \mu\text{m}$, cladding $125\ \mu\text{m}$. MMGI: core $50\text{--}62.5\ \mu\text{m}$, cladding $125\ \mu\text{m}$.

Q11. State the wave equation for the electric field in a dielectric medium.

$\nabla^2 \vec{E} = \mu\epsilon \partial^2 \vec{E} / \partial t^2$. The speed of propagation is $v = 1/\sqrt{\mu\epsilon} = c/n$.

Q12. What is the Poynting vector and what does it represent in fiber optics?

$\vec{S} = \vec{E} \times \vec{H}$ (W/m^2). It represents the direction and magnitude of electromagnetic power flow. In a fiber, it points along the fiber axis for guided modes.

Summary: Fiber Type Comparison at a Glance

Feature	SMF	MMSI	MMGI
V-number	< 2.405	Large	Large
Core dia.	8–10 μm	50–200 μm	50–62.5 μm
Dispersion	Lowest	Highest	Low
Bandwidth	Highest	Lowest	Medium
Source	Laser	LED/Laser	LED/Laser
Use case	Long haul	Short links	LAN/campus